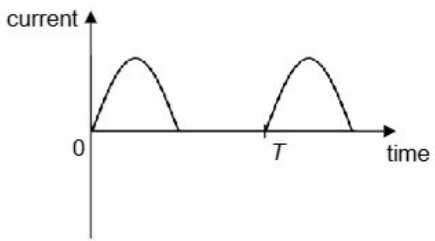
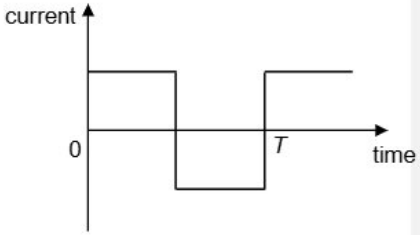
21 A half wave rectified sinusoidal and a square wave current are passed separately through the same resistor. In each case, the resistor dissipates the same mean power.



half wave rectified sinusoidal wave



square wave

What is the ratio $\frac{\text{peak current of square wave}}{\text{peak current of half wave rectified sinusoidal wave}}$?

A $\frac{1}{2}$

B $\frac{1}{\sqrt{2}}$

C 1

D 2